Data Analytics Course

Program Outline

This Data Analytics course equips learners with a comprehensive skill set to analyze, visualize, and interpret data for actionable insights. The curriculum begins with the foundations of data analytics, career pathways, and tools, before diving into Excel (including AI-powered features), statistics, SQL, ETL, Power BI, Tableau, Python, EDA, machine learning and Gen AI in DA. Learners progressively master essential tools, programming, and business intelligence platforms, preparing them to handle end-to-end analytics workflows.

The course emphasizes applied learning, from statistical foundations and SQL querying to modern visualization tools and cloud-enabled AI assistants. With hands-on projects in Excel Copilot, Power BI AI, Tableau AI, and machine learning, learners build practical, job-ready expertise for roles in business intelligence, reporting, and data analysis.

Learning Objectives

- Understand the fundamentals of data analytics, career pathways, and industry tools.
- Gain strong proficiency in Microsoft Excel, including functions, dashboards, macros, and AI-powered Copilot.
- Apply descriptive and inferential statistics to real-world problems, including probability distributions, hypothesis testing, and ANOVA.
- Develop advanced SQL querying skills, including joins, subqueries, CTEs, and window functions.
- Learn ETL foundations to extract, clean, transform, and load data into analytical tools.
- Build interactive reports and dashboards in Power BI and Tableau, enhanced with AI features.
- Acquire hands-on Python programming skills for data manipulation, visualization, and analysis.
- Implement machine learning models including regression, classification, and evaluation metrics for predictive analytics.
- Strengthen data storytelling and reporting skills with stakeholder-ready dashboards.

Learning Outcomes

By the end of this course, learners will be able to:

- Use Excel and Excel Copilot for data preparation, automation, and AI-driven analysis.
- Apply statistical concepts to data interpretation, hypothesis testing, and decision-making.
- Write efficient SQL queries with advanced database techniques for analytics.
- Build ETL pipelines for structured data workflows.
- Develop professional reports and dashboards in Power BI and Tableau with AI integration.
- Use Python (NumPy, Pandas, Matplotlib, Seaborn, Plotly) for data manipulation and visualization.
- Perform exploratory data analysis (EDA) on real-world datasets.
- Implement machine learning models (linear regression, logistic regression, regularization, classification).
- Deliver end-to-end analytics projects with reporting, visualization, and presentation.

Tools & Technologies

- Excel & Excel Copilot
- Power BI (with AI features)
- Tableau (with AI Pulse)
- SQL Databases (MySQL, PostgreSQL)
- **Python:** NumPy, Pandas, Matplotlib, Seaborn, Plotly
- **Python IDEs & Notebooks:** Jupyter Notebook, VS Code, PyCharm, Google Colab, Deepnote.
- **Data Structures & Operations:** NumPy, Pandas.
- **Interactive Dashboards:** Power BI, Tableau, Excel.
- **File Handling & Data Storage:** CSV, JSON, Excel
- ETL Tools (Excel, SQL, automation tools)
- **Visualization:** Dashboards, KPI reports, drill-down navigation
- **Machine Learning:** scikit-learn
- **Generative AI tools:** Co-pilot, PowerBI AI features, Tableau Pulse, Microsoft Fabric, NLPAug.

Curriculum

Milestone 1: Introduction to Data Analytics | Week 1

- What is Data Analytics?
- Career Pathways and Growth Roadmap
- What does it take to be a Data Analyst?
- Tools to Get Started
- Types of Data Analytics: Descriptive, Diagnostic, Predictive, Prescriptive

Milestone 2: Excel with AI | Week 2–5

MS Excel Basics and Functions

- Installation (MS Office, Web version, Microsoft 365 trial)
- Excel Interface & Navigation (Part 1 & 2)
- Data Entry, Editing, Formatting, Clipboard, Split Panes
- Insert/Delete Rows, Columns, Cells
- Freeze Panes, Hide/Unhide, Worksheets (Add, Delete, Rename, Group, Ungroup)
- Basic Formulas and Predefined Functions
- Arithmetic, Logical, Statistical, Database Functions
- XLOOKUP, INDEX-MATCH, OFFSET

Excel Functions & Advanced Data Handling

- Cell Referencing (Absolute, Relative, Mixed)
- Mathematical, Text, Logical, Date-Time Functions
- VLOOKUP, HLOOKUP, LOOKUP
- Tables, Sorting, Filtering, Handling Duplicates
- Scenario Manager, Goal Seek

Excel Dashboards, Macros & Data Management

- What-if Analysis, Consolidation
- Conditional Formatting
- Pivot Tables, Slicers, Charts Formatting
- Dashboard Creation
- Macros & VBA: Recording, Editing, Loops, Input Boxes
- Worksheet & Workbook Protection

Excel with Copilot

- Introduction to Copilot & Access
- Data Preparation & Cleaning with AI Assistance
- Natural Language Queries
- Formula Writing & Debugging
- Creating Charts & Dashboards from Prompts
- Forecasting & Advanced Analysis
- Automating Reports with AI

Milestone 3: Mastering Statistics | Week 6–8

Fundamentals of Statistics

- Types of Statistics, Sampling, Data Types, Levels of Measurement
- Measures of Central Tendency, Variance, Standard Deviation, Skewness
- Covariance & Correlation

Probability Distributions & Statistical Inference

- Random Variables, PMF, PDF, CDF
- Distributions: Uniform, Bernoulli, Binomial, Poisson, Normal
- Central Limit Theorem, Z-Statistics, Estimates

Advanced Statistical Methods

- Hypothesis Testing: Z, T, Chi-square, F-tests, ANOVA
- Errors (Type I & II), p-value, Confidence Intervals
- Bayes' Theorem, Goodness-of-fit tests

Milestone 4: Mastering SQL | Week 9–10

SQL Fundamentals

- Installation & Setup
- SQL Commands (DML, DQL)
- Clauses (WHERE, ORDER BY, LIMIT, OFFSET)
- Constraints & Operators

Advanced SQL Queries & Database

- IN, LIKE, GROUP BY, String Functions, Date-Time Functions
- CTE, Window Functions, Numeric Functions
- Joins & Subqueries
- Indexing, Normalization, ER Diagrams

- SQL with GPT integration

Project

Project	Description
Flight Data Analysis & Operational Optimization for Airports	Working on airport data to analyse flight data and identify factors which can optimize operations.

Milestone 5: ETL Foundations | Week 11–12

- Introduction to ETL for Data Analysts
- Data Extraction from Multiple Sources
- Cleaning, Formatting, Standardization
- Transformation using SQL & Excel
- Loading into Analytical Tools
- Automating ETL Processes
- Overview of Common ETL Tools

Milestone 6: Mastering Power BI with AI | Week 13–15

Power BI Intro & Charts

- Installation, User Interface
- Visualizations: Bar, Column, Pie, Donut, Waterfall, Treemap, Table, Matrix, Line, Area, KPI, Funnel, Scatter, Maps, Gauge, Card, Custom Visuals
- Data Modeling & Relationships, Schemas, Optimizations

DAX & Creating Reports

- DAX Basics (Row/Filter Context, Measures, Calculated Columns)
- Functions: SUM, SUMX, MIN, MAX, AVERAGE, COUNT, DISTINCTCOUNT
- Advanced DAX: FILTER, CALCULATE, CROSSFILTER, Date, String, Logical
- Report Creation, Power BI Service

Power BI AI Features

- Key Drivers with Influencers
- Decomposition Tree
- Q&A Visual
- Anomaly Detection & Trend Analysis

- AI in Power Query

Project

Project	Description
Superstore Sales Analysis	Working on superstore dataset to analyse and create an insightful dashboard which can help businesses grow.

Milestone 7: Mastering Tableau with AI | Week 16–19

Introduction to BI & Tableau

- BI Tools & Methodologies
- Tableau Architecture & Products
- Tableau Prep & Installation
- Connecting to Data Sources

Applied Data Visualization

- Bar Charts, Treemaps, Pie Charts, Text Tables, Maps
- Heatmaps, Line Charts, Word Clouds, Histograms, Scatter Plots

Advanced Tableau Techniques

- Gantt, Waterfall, Reference Lines, Dual Axes, Animated Charts
- Funnel, Dumbbell, Butterfly, Box-Whisker
- Filters, Sets, Parameters, Calculations, Joins, Data Blending
- Dashboards, Tableau Server, User Security, Scheduling

Tableau AI Pulse

- AI-Generated Summaries
- Metric Alerts & Trusted Analytics
- Q&A with Metrics
- Drill-Downs & Correlation Insights

Milestone 8: Reporting Essentials | Week 20

- Report Requirements & KPIs
- Visual Selection & Storytelling
- Drill-down & Drill-through Navigation

- Mobile-Friendly Reports
- Theming, Branding, Performance Optimization
- Stakeholder Validation

Milestone 9: Python for Data Analytics | Week 21–22
Python Basics

- Keywords, Mutability, Operators, Type Casting
- Conditionals & Loops

Data Types & Structures

- Strings, Lists, Tuples, Sets, Dictionaries

Functions

- Iterators & Generators
- Lambda, Map, Reduce, Filter

Data Toolkit

- NumPy (Basic & Advanced)
- Pandas (Basic & Advanced)
- Visualization: Matplotlib, Seaborn, Plotly

Milestone 10: Exploratory Data Analysis (EDA) | Week 23

- Flight Price Prediction Case Study
- US Visa Dataset Analysis

Milestone 11: Machine Learning for Data Analytics | Week 24–25
Regression

- Simple & Multiple Linear Regression
- Polynomial Regression
- Feature Importance, R^2, Adjusted R^2
- MSE, MAE, RMSE
- ML Pipelines & Model Pickling

Classification

- Logistic Regression, Multicollinearity
- Regularization (Lasso, Ridge, Elastic Net)

- Cross-validation & Hyperparameter Tuning
- Classification Metrics
- Multiclass Classification